

GOLDSBORO-WAYNE, NC, USA

WMO: 746940

Lat: 35.460N

Lon: 77.965W

Elev: 134

StdP: 14.62

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 03708

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.0	25.1	4.8	7.1	29.2	9.6	9.0	32.1	20.9	57.9	18.8	53.9	3.8	0	0.391

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	95.1	75.9	92.5	75.8	90.4	75.3	79.9	89.2	78.5	87.3	77.7	86.0	6.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.5	143.6	83.2	76.7	139.7	82.3	75.4	133.7	80.8	43.4	88.5	42.1	86.9	41.1	85.8	91.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.4	16.1	13.3	DB	15.0	99.6	3.1	4.3	12.8	102.7	11.0	105.2	9.3	107.7	7.1	110.8
			WB	12.6	82.3	4.3	2.9	9.5	84.4	7.0	86.1	4.6	87.7	1.5	89.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	61.7	42.3	45.2	51.9	61.6	70.0	77.6	80.2	78.1	72.6	62.4	51.8	46.2	(6)
(7)		DBStd	15.40	10.57	9.41	9.54	8.23	7.06	5.34	3.72	4.09	5.17	7.92	8.57	9.17	(7)
(8)		HDD50	854	288	191	96	8	0	0	0	0	6	78	187		(8)
(9)		HDD65	3017	706	557	417	160	36	1	0	7	144	404	585		(9)
(10)		CDD50	5141	48	57	154	356	619	827	936	871	678	391	133	71	(10)
(11)		CDD65	1824	1	3	11	57	190	377	471	406	234	63	8	3	(11)
(12)		CDH74	16346	1	17	100	542	1697	3683	4600	3475	1697	473	52	9	(12)
(13)		CDH80	6580	0	0	8	117	609	1645	2094	1446	555	102	4	0	(13)
(14)	Wind	WSAvg	5.1	5.6	6.1	6.3	6.5	5.5	4.9	4.4	3.4	4.2	4.4	4.7	4.8	(14)
(15)	Precipitation	PrecAvg	50.1	3.5	3.3	3.7	3.6	3.9	4.8	5.9	5.4	6.1	3.4	3.3	3.1	(15)
(16)		PrecMax	70.1	7.3	7.4	6.1	6.5	8.6	12.9	12.3	14.7	28.6	7.0	7.8	6.6	(16)
(17)		PrecMin	35.7	0.9	1.4	0.9	1.1	1.8	1.5	2.7	1.7	1.1	0.0	0.3	0.6	(17)
(18)		PrecStd	8.3	1.8	1.7	1.5	1.6	1.7	2.6	2.4	2.6	5.3	1.9	2.0	1.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.2	76.9	81.2	87.3	94.5	97.5	98.6	96.6	91.6	87.7	80.5	74.8	(19)
(20)		MCWB	64.2	65.0	63.5	68.1	73.6	74.3	77.7	76.7	73.9	72.8	67.5	67.3		(20)
(21)		2%	DB	69.2	71.4	77.0	83.2	89.9	94.8	95.1	92.6	89.5	82.3	74.6	70.4	(21)
(22)		MCWB	61.9	61.3	62.4	65.8	71.0	74.8	77.2	76.7	73.7	69.5	63.4	64.9		(22)
(23)		5%	DB	63.7	66.6	72.7	80.7	86.5	91.4	92.2	90.3	86.2	79.2	70.8	66.7	(23)
(24)		MCWB	57.4	58.7	60.3	64.7	70.6	74.2	77.1	76.5	73.0	67.5	63.5	63.0		(24)
(25)		10%	DB	59.4	62.2	68.4	77.0	82.8	89.7	90.2	88.2	82.5	75.8	67.6	62.8	(25)
(26)		MCWB	53.8	54.5	58.4	63.7	69.4	73.9	76.4	75.8	71.7	67.8	61.7	58.1		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.3	67.2	67.5	72.9	76.4	79.3	82.0	81.0	78.4	76.7	71.1	70.3	(27)
(28)		MCDWB	69.7	72.9	74.7	78.8	85.7	89.3	91.9	90.1	85.5	80.9	76.0	72.6		(28)
(29)		2%	WB	63.0	64.3	65.4	70.1	74.5	77.9	80.3	79.7	77.1	74.1	68.1	66.5	(29)
(30)		MCDWB	66.7	69.0	72.6	76.5	83.4	87.9	89.9	88.6	84.2	78.3	71.0	69.0		(30)
(31)		5%	WB	59.3	60.5	63.4	68.1	73.0	76.9	79.0	78.4	75.6	71.7	65.2	63.2	(31)
(32)		MCDWB	62.7	65.5	68.8	75.4	81.4	86.5	88.2	86.4	81.3	76.8	69.6	66.1		(32)
(33)		10%	WB	55.1	55.5	61.1	66.0	71.6	75.8	77.9	77.3	74.2	69.2	62.0	59.0	(33)
(34)		MCDWB	58.6	60.4	67.3	74.1	79.6	84.7	86.5	84.6	79.3	73.3	66.6	62.1		(34)
(35)	Mean Daily Temperature Range	MDBR	18.5	20.2	22.1	23.2	21.2	20.3	19.6	18.6	18.9	21.3	22.1	19.3		(35)
(36)		5% DB	MCDBR	23.8	25.5	26.2	26.1	24.1	23.0	22.3	21.3	22.5	23.8	24.6	21.9	(36)
(37)		MCWBR	17.6	17.2	14.0	11.5	8.6	7.4	7.9	8.1	9.3	11.8	15.2	16.4	(37)	
(38)		5% WB	MCDBR	21.1	24.1	21.8	20.9	19.9	19.7	19.9	18.2	17.4	17.6	19.6	20.1	(38)
(39)		MCWBR	18.1	18.8	14.6	11.5	8.3	7.4	8.4	7.9	8.4	10.2	14.6	17.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.315	0.332	0.360	0.406	0.457	0.503	0.537	0.525	0.421	0.370	0.333	0.316	(40)	
(41)		taud	2.492	2.440	2.393	2.268	2.161	2.078	1.999	2.023	2.313	2.433	2.494	2.525	(41)	
(42)		Ebn at Noon	282	289	288	279	265	252	242	242	265	271	272	274	(42)	
(43)		Edh at Noon	26	31	36	43	48	52	56	54	38	31	26	24	(43)	
(44)	All-Sky Solar Radiation	RadAvg	824	1042	1374	1735	1912	2029	1935	1743	1452	1182	907	724	(44)	
(45)		RadStd	54	115	94	115	142	127	111	95	110	151	71	58	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (56 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+292	+203

Nomenclature: See separate page